

USER SERVICE

PORT: 8081

INTRODUCTION

User Service handles Customer, Admin, and Driver registration/login with comprehensive account management, address management, activity tracking, and inter-service communication capabilities.

BACKEND STRUCTURE

Data Base Storage: MySQL

Data Base Name: USERDB

Data Base Tables: users, addresses, token_blacklist, otp, user_activity_logs

CLASS PRINTS AND DB DATA

S.No.	Class	Properties/Attributes	Table name	Column names DB
1	User	Long id String name String email String mobile String password String roleName String userCode Set<Role> roles String refreshToken LocalDateTime refreshTokenExpiry boolean emailVerified boolean mobileVerified boolean active boolean deleted	users	id name email mobile password role_name user_code refresh_token refresh_token_ex email_verified mobile_verified active deleted
2	Address	Long id Long userId String addressLine1 String addressLine2 String city String state String pincode boolean isDefault LocalDateTime createdAt LocalDateTime updatedAt	addresses	id user_id address_line1 address_line2 city state pincode is_default created_at updated_at
3	TokenBlacklist	Long id String token LocalDateTime expiryTime LocalDateTime createdAt	token_blacklist	id token expiry_time created_at
4	OTP	Long id String identifier String otp OTPPurpose purpose LocalDateTime expiryTime boolean verified LocalDateTime createdAt	otp	id identifier otp purpose expiry_time verified created_at
5	UserActivityLog	Long id Long userId String userEmail String userRole ActivityType activityType String description String ipAddress String userAgent LocalDateTime timestamp	user_activity_logs	id user_id user_email user_role activity_type description ip_address user_agent timestamp

ENUMS

S.No.	Enum	Constants
1	Role	CUSTOMER, ADMIN, DRIVER
2	OTPPurpose	EMAIL_VERIFICATION, MOBILE_VERIFICATION, FORGOT_PASSWORD
3	ActivityType	REGISTRATION, LOGIN, LOGOUT, PROFILE_UPDATE, EMAIL_CHANGE, PASSWORD_CHANGE, ADMIN_ACTION

SERVICE FUNCTIONALITIES BASED ON ROLES AND API END-POINTS

1. ADMIN FUNCTIONALITIES

S.No.	Role	Functionality	Token Requirement	Details/Endpoints
1	Admin	ADMIN can register DRIVERS	Yes	/api/users/register/driver
2	Admin	ADMIN can view all users	Yes	/api/users/all
3	Admin	ADMIN can view user by id	Yes	/api/users/{id}
4	Admin	ADMIN can update its own password only	Yes	/api/users/updateEmployee/{id}
5	Admin	ADMIN can update driver's name/email/vehicle number/phone number only	Yes	/api/users/updateDriverDetails/{id}
6	Admin	Admin can delete any customer's account	Yes	/api/users/deleteAccount/{id}
7	Admin	Admin can soft delete any driver's account	Yes	/api/users/softDeleteEmployee/{id}
8	Admin	Admin can get list of active customers (paginated)	Yes	/api/users/customers/active
9	Admin	Admin can get one customer details	Yes	/api/users/customers/{id}
10	Admin	Admin can activate customer account	Yes	/api/users/customers/{id}/activate
11	Admin	Admin can deactivate customer account	Yes	/api/users/customers/{id}/deactivat
12	Admin	Admin can generate internal JWT token for microservices	Yes	/api/auth/generate-internal-token
13	Admin	Admin can view user activity history	Yes	/api/analytics/users/{userId}/activit
14	Admin	Admin can view last login time	Yes	/api/analytics/users/{userId}/last-login
15	Admin	Admin can view login frequency	Yes	/api/analytics/users/{userId}/login-frequency
16	Admin	Admin can view system statistics	Yes	/api/analytics/system/statistics
17	Admin	Admin can view recent activities	Yes	/api/analytics/activities/recent
18	Admin	Admin can view admin action logs	Yes	/api/analytics/activities/admin

2. CUSTOMER FUNCTIONALITIES

S.No.	Role	Functionality	Token Requirement	Details/Endpoints
1	Customer	Anyone can register as CUSTOMER	No	/api/users/register/cus
2	Customer	LoggedIn customer can update name/email/mobile/password	Yes	/api/users/updateCustc
3	Customer	LoggedIn customer can delete its own account	Yes	/api/users/deleteAccou
4	Customer	LoggedIn customer can deactivate its own account	Yes	/api/users/customer/de
5	Customer	Customer can add multiple addresses	Yes	/api/addresses
6	Customer	Customer can view all saved addresses	Yes	/api/addresses
7	Customer	Customer can update address	Yes	/api/addresses/{id}
8	Customer	Customer can delete address	Yes	/api/addresses/{id}
9	Customer	Customer can set default delivery address	Yes	/api/addresses/{id}/se
10	Customer	Customer can send email verification OTP	Yes	/api/auth/verify-email/
11	Customer	Customer can verify email with OTP	Yes	/api/auth/verify-email/otp

3. DRIVER FUNCTIONALITIES

S.No.	Role	Functionality	Token Requirement	Details/Endpoints	Status
1	Driver	Driver can update its own password only	Yes	/api/users/updateEmployee/{id}	Done

4. COMMON FUNCTIONALITIES (Admin, Customer, Driver)

S.No.	Role	Functionality	Token Requirement	Details/Endpoints	Status
1	All	Only registered user can log in, then only token will be generated	No	/api/auth/login	Done
2	All	User can logout (token blacklisting)	Yes	/api/auth/logout	Done
3	All	User can refresh access token	No	/api/auth/refresh	Done
4	All	User can request forgot password OTP	No	/api/auth/forgot-password	Done
5	All	User can reset password with OTP	No	/api/auth/reset-password	Done

5. INTERNAL MICROSERVICE APIs

S.No.	Functionality	Token Requirement	Details/Endpoints	Status
1	Get user basic info by userId	Yes (Internal JWT)	/internal/users/{userId}	Done
2	Validate token and role	Yes (Internal JWT)	/internal/validate-token	Done
3	Check if user can place order	Yes (Internal JWT)	/internal/users/{userId}/can-place-order	Done

TEST CASES

1. Register Customer

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
1	Register Customer	id name email mobile userCode roleName roles	No	name, email, mobile, password present name blank Email blank Email format wrong Email should not contain @meradesh.com Password blank Password format wrong provided email id must not already exist provided mobile number must not already exist	User details, with status 201 CREATED message: "name is required" message: "email is required" message: "invalid email" message = "Customer email cannot use @meradesh.com domain" message: "password is required" message = "password must be at least 8 characters" "Customer already exists with email:" + dto.getEmail() "Mobile number already in use:" + dto.getMobile()	Done Done Done Done Done Done Done

2. Login

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
2	Login					
	email password	token refreshToken user details	No	valid email and password	JWT token, refresh token, user details with status 200 OK	Done
				invalid email or password	message: "Invalid Email or Password"	Done
				user account is deactivated	message: "Account is deactivated. Please contact support."	Done
				user account is deleted (non-driver)	message: "Account has been deleted. Please contact support."	Done

3. Logout

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
3	Logout					
	N/A	message	Yes	valid token provided	"Logged out successfully" with status 200 OK	Done
				no token provided	"No token provided"	Done
				token already blacklisted	"Token has been invalidated"	Done

4. Refresh Token

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
4	Refresh Token					
	refreshToken	token refreshToken	No	valid refresh token	new access token with status 200 OK	Done
				invalid or expired refresh token	message: "Invalid or expired refresh token"	Done

5. Forgot Password

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
5	Forgot Password					
	email	message	No	valid email exists in system	"OTP sent to your email" with status 200 OK	Done
				email not found	"User not found with email:" + email	Done

6. Reset Password

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
6	Reset Password					
	email otp newPassword	message	No	valid email, OTP, and password	"Password reset successfully" with status 200 OK	Done
				invalid or expired OTP	"Invalid or expired OTP"	Done
				password too short	"password must be at least 8 characters"	Done

7. Get List of All Users

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
7	Get list of all users					
	N/A	List<> [{} , {}]	Yes	Only ADMIN can get list of all users	List of User details, with status 200 OK	Done
				Not admin	Access Denied	Done
				If no data present in DB	message="No users found in the system"	Done

8. Get User by ID

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
8	Get user by ID					
	id	user details	Yes	Only ADMIN can get user by id	User details, with status 200 OK	Done
				Not admin	Access Denied	Done
				If no user exist with given user id in DB	"User not found with ID:" + id	Done

9. Customer Updating Themselves

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
9	Customer updating themselves (can update name, email, mobile and password)	id name email mobile userCode roleName roles	Yes	Only LOGGED In CUSTOMER can change his/her name, email, mobile or password	Updated user details, with status 200 OK	Done
				Other Logged in trying to update others field	"You can update only your own account."	Done
				Email format wrong	message: "invalid email"	Done
				Email should not contain @meradesh.com	message = "Customer email cannot use @meradesh.com domain"	Done
				Password format wrong	message = "password must be at least 8 characters"	Done
				If no user exist with given user id in DB	"User not found with ID:" + id	Done
				Newly requested email id must not be in use with other user	"Email already in use:" + dto.getEmail()	Done
				Newly requested mobile must not be in use with other user	"Mobile number already in use:" + dto.getMobile()	Done

10. Get Active Customers (Admin Only)

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
10	Get Active Customers					
	page size	Page<User>	Yes	Only ADMIN can get active customers	Paginated list of active customers with status 200 OK	Done
				Not admin	Access Denied	Done

11. Activate Customer (Admin Only)

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
11	Activate Customer					
	customerId	message	Yes	Only ADMIN can activate customer	"User activated successfully" with status 200 OK	Done
				Not admin	Access Denied	Done
				User not found	"User not found with ID:" + id	Done
				User is not a customer	"Only customer accounts can be activated via this endpoint"	Done

12. Deactivate Customer (Admin Only)

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
12	Deactivate Customer					
	customerId	message	Yes	Only ADMIN can deactivate customer	"User deactivated successfully" with status 200 OK	Done
				Not admin	Access Denied	Done
				User not found	"User not found with ID:" + id	Done
				User is not a customer	"Only customer accounts can be deactivated via this endpoint"	Done

13. Customer Self-Deactivate

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
13	Customer Self-Deactivate					
	N/A	message	Yes	Logged in customer deactivates own account	“Your account has been deactivated” with status 200 OK	Done
				Not logged in	Access Denied	Done

14. Add Address

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
14	Add Address					
	addressLine1 addressLine2 city state pincode	id addressLine1 addressLine2 city state pincode isDefault createdAt updatedAt	Yes	All required fields present	Address details with status 201 CREATED	Done
				addressLine1 blank	message: “addressLine1 is required”	Done
				city blank	message: “city is required”	Done
				state blank	message: “state is required”	Done
				pincode blank	message: “pincode is required”	Done
				pincode not 6 digits	message: “Pincode must be 6 digits”	Done
				First address added	isDefault automatically set to true	Done

15. Update Address

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
15	Update Address addressId addressLine1 addressLine2 city state pincode	updated address details	Yes	User updates own address User tries to update another user's address Address not found	Updated address details with status 200 OK "Address not found" with status 404 "Address not found" with status 404	Done Done Done

16. Delete Address

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
16	Delete Address addressId	message	Yes	User deletes own address User tries to delete another user's address Deleted address was default	"Address deleted successfully" with status 200 OK "Address not found" with status 404 Another address automatically becomes default	Done Done Done

17. Set Default Address

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
17	Set Default Address	updated address details	Yes	User sets own address as default	Address with isDefault=true, status 200 OK	Done
				Previous default address	Previous default automatically unset	Done
				User tries to set another user's address	"Address not found" with status 404	Done

18. Email Verification

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
18	Send Email Verification OTP	message	Yes	User requests email verification OTP	"Verification OTP sent to your email" with status 200 OK	Done
				Email already verified	"Email already verified"	Done
19	Verify Email with OTP	message	Yes	Valid OTP provided	"Email verified successfully" with status 200 OK	Done
				Invalid or expired OTP	"Invalid or expired OTP"	Done
				Email already verified	"Email already verified"	Done

19. Generate Internal JWT Token (Admin Only)

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
20	Generate Internal JWT Token	serviceName internalToken tokenType expiresIn usage allowedEndpoints	Yes	Only ADMIN can generate internal token Not admin serviceName blank	Internal JWT token details with status 200 OK Access Denied "Service name is required"	Done Done Done

20. Get User Activity History (Admin Only)

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
21	Get User Activity History	Page<UserActivityLog>	Yes	Only ADMIN can view activity history Not admin	Paginated activity logs with status 200 OK Access Denied	Done Done

21. Get System Statistics (Admin Only)

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
22	Get System Statistics	totalLogins totalRegistrations loginsToday registrationsToday loginsThis Week registrationsThis Week	Yes	Only ADMIN can view statistics Not admin	System statistics with status 200 OK Access Denied	Done Done

22. Internal API - Get User Info

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Status
23	Get User Info (Internal)	id name email userCode roles active deleted emailVerified mobileVerified	Yes (Internal JWT)	Valid internal JWT token	User info with status 200 OK	Done
				No internal JWT token	“Missing or invalid internal token” with status 401	Done
				User not found	“User not found” with status 404	Done

23. Internal API - Check Order Eligibility

S.No.	Request DTO	Response DTO	Token	Case Input	Case Output	Sta
24	Check Order Eligibility (Internal)	userId canPlaceOrder emailVerified active deleted reason	Yes (Internal JWT)	User is eligible (email verified, active, not deleted)	canPlaceOrder=true with status 200 OK	Doi
				Email not verified	canPlaceOrder=false, reason=“Email not verified”	Doi
				Account deactivated	canPlaceOrder=false, reason=“Account is deactivated”	Doi
				Account deleted	canPlaceOrder=false, reason=“Account is deleted”	Doi

CONFIGURATION REQUIREMENTS

Gmail SMTP Configuration

For email functionality (OTP, verification), configure Gmail credentials in application.properties:

```
spring.mail.username=your-email@gmail.com
spring.mail.password=your-gmail-app-password
```


How to generate Gmail App Password: 1. Go to Google Account Settings 2. Security > 2-Step Verification (enable if not enabled) 3. Security > App Passwords 4. Generate password for “Mail” 5. Use the 16-character password in application.properties

Internal JWT Configuration

For microservice communication, configure internal JWT secret:

```
internal.jwt.secret=your-secure-secret-key  
internal.jwt.expiration=3600000
```

PENDING FEATURES

S.No.	Feature	Status	Notes
1	Mobile OTP Verification	Pending	Infrastructure ready, awaiting SMS service provider integration

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Last Updated: January 28, 2026

Total Endpoints: 40+

Status: Production Ready